CBSE Test Paper 05

Chapter 9 Mechanical Properties of Solids

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1. In a materials testing laboratory, a metal wire made from a new alloy is found to break when a tensile force of 90.8 N is applied perpendicular to each end. If the diameter of the wire is 1.84 mm, what is the breaking stress of the alloy? **1**
   a. 3.41×10^7 Pa
   b. 3.61×10^7 Pa
   c. 3.31×10^7 Pa
   d. 3.51×10^7 Pa
2. Compute the bulk modulus of water from the following data: Initial volume = 100.0 litre, Pressure increase = 100.0 atm (1 atm = 1.013×10^5 Pa), Final volume = 100.5 litre **1**
   a. 2.226×10^9 Pa
   b. 2.126×10^9 Pa
   c. 2.326×10^9 Pa
   d. 2.026×10^9 Pa
3. The edge of an aluminum cube is 10 cm long. One face of the cube is firmly fixed to a vertical wall. A mass of 100 kg is then attached to the opposite face of the cube. The shear modulus of aluminum is 25 GPa. What is the vertical deflection of this face? **1**
   a. 4.0×10^{-7} m
   b. 6.0×10^{-6} m
   c. 8.0×10^{-6} m
   d. 2.0×10^{-6} m
4. A perfectly rigid body is one **1**
   a. which does not move on application of force
   b. whose shape and size do not change on application of force
   c. which starts flowing like water on application of force
   d. whose shape and size change on application of force
5. What is the density of water at a depth where pressure is 80.0 atm, given that its density at the surface is 1.03×10^3 kg m^{-3}? **1**
   a. 1.054×10^3 kg/ m^3

b. 1.074×10^3 kg/ m^3
c. 1.094×10^3 kg/ m^3
d. 1.034×10^3 kg/ m^3

6. Is stress a vector quantity? **1**
7. Is it possible to double the length of a metallic wire by applying a force over it? **1**
8. How are we able to break a wire by repeated bending? **1**
9. The Young's modulus for steel is much more than that for rubber. For the same longitudinal strain, which one will have greater tensile stress? **2**
10. The length of a metal is l_1, when the tension in it is T_1 and is l_2 when tension is T_2. Find the original length of wire? **2**
11. Which is more elastic rubber or steel? Explain. **2**
12. Explain the following: **3**
    i. Concrete beams used in large buildings have greater depth than breadths.
    ii. Load bearing bars are generally made in I-section.
    iii. Pillar with distributed ends is preferred over a pillar with rounded ends.
13. A force of 5×10^3 N is applied tangentially to the upper face of a cubical block of steel of side 30 cm. Find the displacement of the upper face relative to the lower one, and the angle of shear. The shear modulus of steel is 8.3×10^{10} pa. **3**
14. The Marina trench is located in the Pacific Ocean, and at one place it is nearly eleven km beneath the surface of water. The water pressure at the bottom of the trench is about 1.1×10^8 Pa. A steel ball of initial volume 0.32 m^3 bis dropped into the ocean and falls to the bottom of the trench. What is the change in the volume of the ball when it reaches to the bottom? (Bulk modulus of steel, k = 1.6×10^{11} N/m^2) **3**
15. A 14.5 kg mass, fastened to the end of a steel wire of unstretched length 1.0 m, is whirled in a vertical circle with an angular velocity of 2 rev/s at the bottom of the circle. The cross-sectional area of the wire is 0.065 cm^2. Calculate the elongation of the wire when the mass is at the lowest point of its path. **5**